University of Central Punjab
Faculty of Information Technology
Linear Algebra
Final Term Exam

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Course Code: CSSS2753, SESS2743	Marks: 50
Duration : 150 minutes	Section:
Date : 13 July 2022	Program: BSCS, BSSE

Name: Registration Number:

Instructions

1. Write your name and registration number on the answer sheet and sign it before submission.
2. Attempt all questions.
3. All your calculations should be on your answer sheet.
4. **ANSWER ALL QUESTIONS IN THE CORRECT ORDER.**

Question 1: (3+4+3 marks)

A toy shop has 3 types of superhero action figures: Spider-man, Hulk, and Wonder-woman.
The spider-man toy costs \$4, the bat-man toy costs \$10, and the wonder-woman action figure toy costs \$12.
Mr. Robert wants to spend exactly \$60 to buy 8 action-figure toys for his nephews and nieces.
He desires to buy at least one type of each action figure toy.
How many action figures of each type can he buy?

a. Set up a system of equations and associated augmented matrix to find the number of each type of action figure.
b. Solve the system to find the parametric solution using the Gauss Elimination method.
c. Use the parametric solution to determine the valid/unique solution for the number of each type of toy.

Question 2: (5+5 pts)

Consider the triangle ΔABC with vertices $\mathbf{A(1,1)}$, $\boldsymbol{B}(\mathbf{4},\mathbf{0})$ and $\boldsymbol{C}$ $(\mathbf{2},\mathbf{3})$.
Find the vertices of the triangle after shearing it along the y-axis by a factor 2 and then reflecting the sheared triangle in the line $y = x$.
Also, sketch the image of the original and transformed triangles.

Question 3 i) : (6 pts)

Let $\boldsymbol{V} = \boldsymbol{R^3} = \{(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}), \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \boldsymbol{R}\}$ be a set of triples of real number with the following addition and scalar multiplication operations:

$$\overrightarrow{u_1} + \overrightarrow{u_2} = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 + 1\,, y_1 + y_2 - 1, z_1 + z_2)$$
$$k(x_1, y_1\,, z_1) = (0, ky_1, kz_1).$$

a. Compute $\overrightarrow{u_1} + \overrightarrow{u_2}$ if $\overrightarrow{u_1} = (4, -1,\ 8)$, $\overrightarrow{u_2} = (3, 2, -5)$.
b. Compute $\overrightarrow{ku_1}$ if $\overrightarrow{u_1} = (4, -1,\ 8)$ and $k = 2$.
c. Find the zero vector of the set.
d. Find the additive inverse of the vector $(2, 3, 4)$.
e. Prove or disprove the property: $k(\overrightarrow{u_1} + \overrightarrow{u_2}) = k\overrightarrow{u_1} + k\overrightarrow{u_2}$.
f. Prove or disprove the property: $1\vec{u} = \vec{u}$

ii) [4 pts] Let $\boldsymbol{V} = \boldsymbol{R^2} = \{(\boldsymbol{x}, \boldsymbol{y}),\ \boldsymbol{x} \in \boldsymbol{R}, \boldsymbol{y} \in \boldsymbol{R}\}$ be a vector space under standard addition and scalar multiplication.
Prove/Disprove the following subsets are ***subspaces***.

a. $W_1 = \{(x, y): x \in R, y = -2x\}$
b. $W_2 = \{(x, y): x = y^2, y \in R\}$

Question 4: [4+6pts]

i) Let $P_2(x)$ be a vector space of all 2nd degree polynomials.
Show that $p = x^2 - 1$, $q = x^2 + x + 2$ and $r = -4x^2 - 2x - 2$ are *linearly dependent* vectors of $P_2(x)$.

ii) Show that the set of vectors $\boldsymbol{S}$ =**{(1,2,3), (1,4,6), (2,-3,-5)}** span R^3.
Use your result to express $v = (9, 1, 0)$ as a linear combination of the given vectors.

Question 5: [5+5 pts] Find all the Eigenvalues of the given matrix. Also, find Eigenvector for ***negative*** eigenvalues for the matrix

$$A = \begin{bmatrix} -2 & 0 & 2 \\ -2 & 1 & 2 \\ 4 & 0 & 5 \end{bmatrix}$$
